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Nuclear Reactors Group

Hybrid Data Assimilation Methods: Application to the DYNASTY Experimental Facility

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7 September 2022

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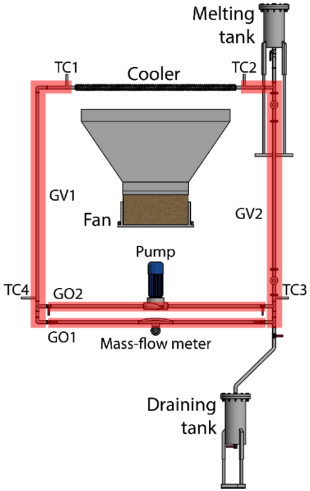
Scope of the work



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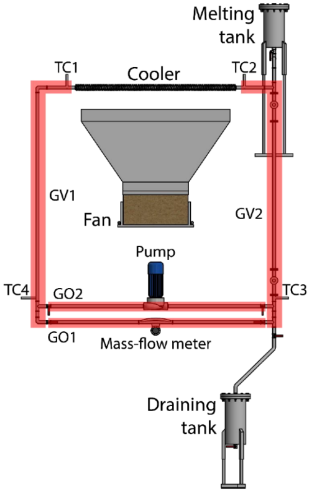
Scope of the work

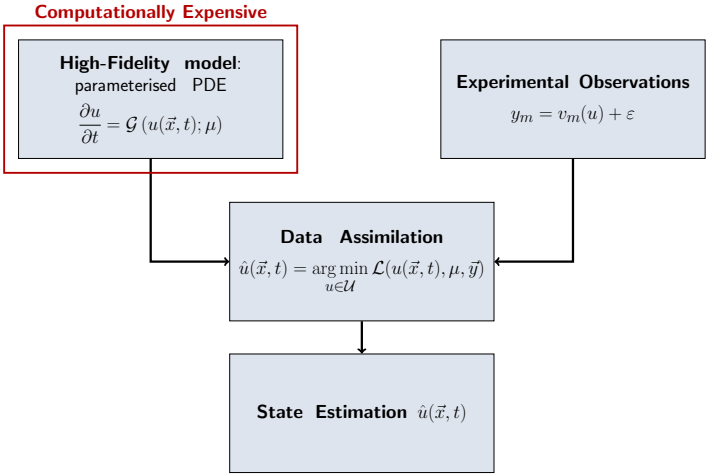


DYNASTY is an experimental facility, aimed to study the natural circulation regime and to simulate the primary circuit of Molten Salt Fast Reactors (MSFRs), with internal heat generation.

Given this experimental facility our aim consists in

> Reconstructing in **real-time** the **full state** of the system (e.g., temperature, velocity and pressure) combining numerical **simulations** and **experimental** observations

Data Assimilation for real-time state estimation



High-Fidelity models are solved using standard numerical methods, such as FE or FV.
The computational cost is typically too heavy for **real-time** applications.

Data Assimilation for real-time state estimation

Computationally Expensive

High-Fidelity model: parameterised PDE

$$\frac{\partial u}{\partial t} = \mathcal{G}\left(u(\vec{x}, t); \mu\right)$$

Experimental Observations

$$y_m = v_m(u) + \varepsilon$$

Data Assimilation

$$\hat{u}(\vec{x}, t) = \arg\min_{u \in \mathcal{U}} \mathcal{L}(u(\vec{x}, t), \mu, \vec{y})$$

State Estimation $\hat{u}(\vec{x}, t)$

High-Fidelity models are solved using standard numerical methods, such as FE or FV.
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Reduced Basis Methods:

$$u(\vec{x}, t; \mu) \simeq \sum_n \alpha_n(t, \mu)\zeta_n(\vec{x})$$

Model Order Reduction via reduced basis methods

The solutions of the mathematical model can be approximated by an expansion on a reduced basis, able to catch the spatial behaviour of the functions

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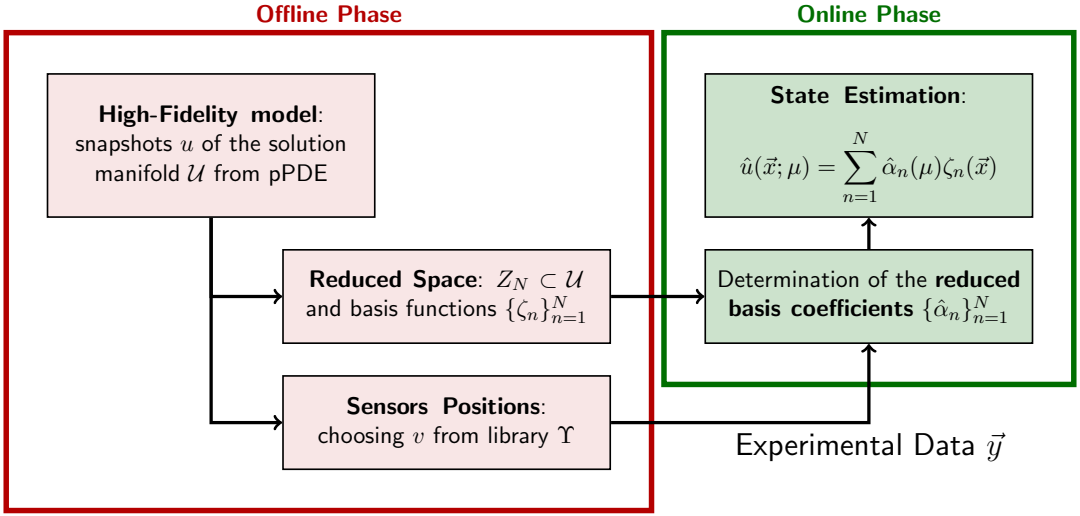
The generation of basis functions is typically computationally demanding, whereas the coefficients can be generated using two main approaches:

- **Intrusive**: a reduced order model (typically a systems of ODEs) is derived from the *high-fidelity* model, e.g. POD-projection methods;
- **Non-intrusive**: the knowledge of the governing equations is not needed. Usually a linear system of small dimension is solved, e.g. EIM or POD-I.

What is the general scheme of HDA techniques?



Description of the techniques

Algorithm 1: Generalised Empirical Interpolation Method (GEIM)

Input: Snapshots $\{u_n\}_{n=1}^N \in \mathcal{U}$

Offline Phase:

$q_m,\, v_m =$ Greedy[a](u_n);

Definition of $B \in \mathbb{R}^{M\times M}$.

Online Phase:

Collect data $\vec{y}$;

$B\vec{\beta} = \vec{y}$;

$\hat{u} \simeq \sum_m \beta_m q_m$.

[a] Grepl et al. (2007)

Algorithm 2: Parameterised-Background Data-Weak (PBDW) formulation

Input: Snapshots $\{u_n\}_{n=1}^N \in \mathcal{U}$

Offline Phase:

$\zeta_m =$ WeakGreedy[b](u_n);

$v_m =$ Greedy(u_n);

Definition of $P \in \mathbb{R}^{(N+M)\times(N+M)}$.

Online Phase:

Collect data $\vec{y}$;

$P\vec{\gamma} = [\vec{y},\, 0]^T$;

$\hat{u} \simeq \sum_m \gamma_m \zeta_m$.

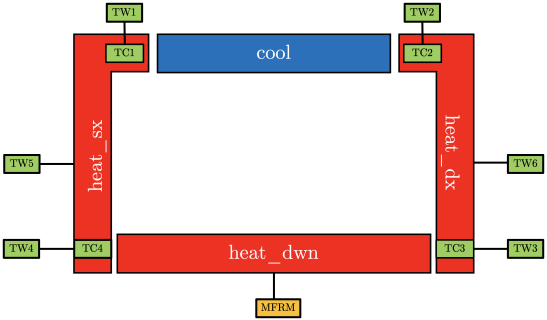
[b] Prud'homme et al. (2002)

Mathematical Modelling

The fluid (water) is heated through electrical stripes, to mimic the internal heat generation, and cooled down with a fan.

Governing equations:

- *Fluid region*: Navier-Stokes + energy equations
- *Solid region*: Heat diffusion equation



Training of the methods

- The field studied is the temperature T with the time t as the only varying parameter μ of the system: the following train and test set are used

$$\Xi_{train} = [0 : 10 : 2900], \qquad \Xi_{test} = [5 : 10 : 2895]; \tag{2}$$

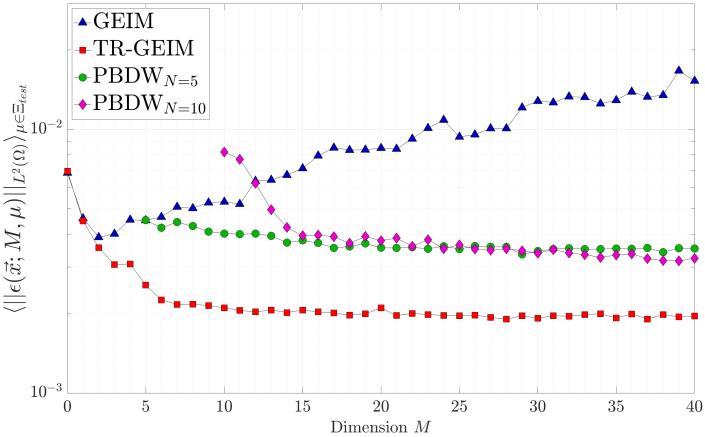
- (TR-)GEIM and PBDW are analysed, considering only synthetic data, polluted with random noise $\varepsilon \sim \mathcal{N}(0, 1)$;
- The error function ϵ is defined as

$$\epsilon(\vec{x}; M, t) = \frac{T^{true}(\vec{x}; t) - \hat{T}(\vec{x}; M, t)}{||T^{true}(\vec{x}; t)||_{L^2(\Omega)}}; \tag{3}$$

- The reduced basis functions and the sensors are selected with the aforementioned methods.

Comparison of the algorithms - Error Plot

- **GEIM**: the method is clearly unstable;
- **TR-GEIM**: the method is able to stabilize the procedure, providing optimal results in reconstructing the temperature field;
- **PBDW**: the formulation provides almost optimal results, even though a lot of sensors may be required.



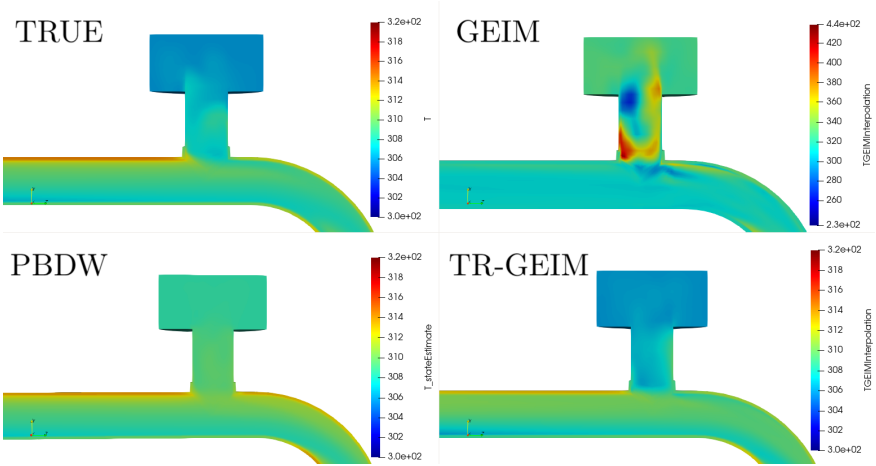
Comparison of the algorithms - Contour Plots

Time $t = 500\,s$;

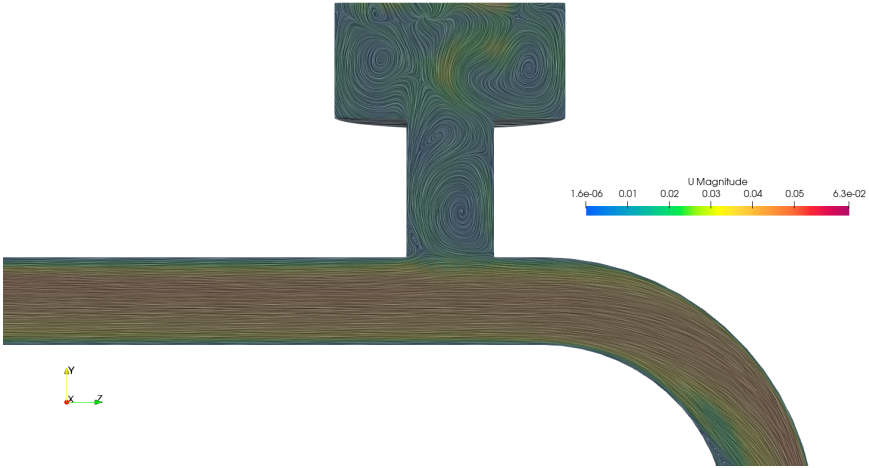
Magic sensors $M = 20$;

Magic functions $M = 20$;

PBDW basis functions $N = 5$.



Analysis of the velocity at the *chimney* region



Final remarks and future developments

(TR-)GEIM and PBDW are promising tools for the integration of MOR methods in a DA framework, i.e. Hybrid Data Assimilation methods. This work has assessed that they are mature enough to be applied to real complex experimental facilities.

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Until now, the model has been considered **perfect** and **only synthetic data** have been considered:

- If **the model is inaccurate**, how will these methods work?
- How will **real experimental data** impact on the reconstruction through HDA?
- How complex flow patterns can be reconstructed?